

Models with Endogenous Regressors

Lecture 6

1 Basics

Suppose that the true model is,

$$y = X\beta + u \tag{1}$$

If $\mathbb{E}(u|X) = 0$, it can be shown that the OLS estimator,

$$\hat{\beta} = \beta + (X'X)^{-1}X'u$$

is unbiased and consistent. Consistency can be shown by proving that the second term converges to zero in probability. First write the second term as

$$(n^{-1}X'X)^{-1}(n^{-1}X'u),$$

and then apply Slutsky theorem, to find that $\hat{\beta} - \beta \rightarrow 0$. However, sometimes even in large samples, the estimator may not be “close” (in some sense) to the true parameter. In the presence of endogenous independent variables (e.g. $\mathbb{E}(u|X) \neq 0$), the estimator $\hat{\beta}$ is biased and inconsistent,

$$\hat{\beta} \rightarrow \beta + Q_{xx}^{-1}Q_{xu} \neq \beta$$

where the matrices Q are limiting forms of $X'X$ and $X'u$. Note that the lack of orthogonality between the error term and the independent variables makes the OLS inconsistent.

1.0.1 Example: Errors in Variables

Many variables (GDP, permanent income, capital depreciation, intelligence, etc.) are measured with errors. Consider the model,

$$y_i = \beta'x_i^* + \varepsilon_i$$

where x_i^* is not observed and independent of ϵ_i . The variable that is observed by the researcher is x_i , which can be expressed as

$$x_i = x_i^* + u_i$$

The OLS estimate of the model,

$$y_i = x_i^{*'}\beta + \varepsilon_i = x_i'\beta + (\varepsilon_i - u_i'\beta) = x_i'\beta + v_i,$$

may be biased and inconsistent because $x \not\perp v$,

$$\mathbb{E}(x_i v_i) = \mathbb{E}(x_i^* + u_i)(\varepsilon_i - u_i'\beta) = -\mathbb{E}u_i u_i'\beta \neq 0$$

Note that

$$\begin{aligned}\mathbb{E}(\hat{\beta}|X) &= (\sum x_i x_i')^{-1} \sum x_i \mathbb{E}(y_i|x_i) \\ &= \beta + (\sum x_i x_i')^{-1} \sum x_i \mathbb{E}(v_i|x_i) \\ &= \beta + \text{bias}\end{aligned}$$

Moreover, in large samples,

$$\begin{aligned}\hat{\beta} &= \beta + (n^{-1} \sum x_i x_i')^{-1} (n^{-1} \sum x_i v_i) \\ &= \beta + (n^{-1} \sum x_i x_i')^{-1} (n^{-1} \sum (x_i^* + u_i)(\varepsilon_i - u_i'\beta)) \\ &= \beta + (n^{-1} \sum x_i x_i')^{-1} (n^{-1} \sum (x_i^* \varepsilon_i - x_i^* u_i' \beta + u_i \varepsilon_i - u_i u_i' \beta))\end{aligned}$$

Since the disturbances ε_i and u_i , and the unobserved variable x_i^* are independent,

$$\hat{\beta} \rightarrow \beta - (Q_{x^*} + Q_u)^{-1} Q_u \beta \neq \beta$$

where $Q_{x^*} = \lim_{n \rightarrow \infty} \sum x_i^* x_i^{*'}$ and $Q_u = \lim_{n \rightarrow \infty} \sum u_i u_i'$. First, note that if one independent variable is measured with error, then, in general, all the estimates are biased and inconsistent (Why?). Second, if $p = 1$ (e.g., one independent variable),

$$\hat{\beta} \rightarrow \frac{\sigma_{x^*}^2}{(\sigma_{x^*}^2 + \sigma_u^2)} \beta$$

suggesting that the greater the measurement error, the worse the bias.

1.0.2 Example: Omitted Variables

The omitting variable bias is created when we do not introduce an unobserved variable. For example, since ability is a latent variable, we cannot estimate the true model returns to education model. On the other hand, there are variables related with the unobserved variable called proxies. For example, the IQ score is a proxy for ability (e.g., the higher person's IQ, the higher the level of ability).

Now, the question is what are the consequences on estimation of using a proxy variable? We consider the return to education model to illustrate and discuss the main lessons.

$$\log(wage_i) = \beta_0 + \beta_1 educ_i + \beta_2 exp_i + \beta_3 abil_i + u_i \quad (2)$$

$$abil_i = \alpha_0 + \alpha_3 IQ_i + \varepsilon_i \quad (3)$$

Replacing equation (6) in (5), we have the following model

$$\log(wage_i) = \delta_0 + \beta_1 educ_i + \beta_2 exp_i + \delta_3 IQ_i + v_i,$$

where $\delta_0 = \beta_0 + \alpha_0\beta_3$, $\delta_3 = \alpha_3\beta_3$, and $v_i = \beta_3\varepsilon_i + u_i$.

By least squares methods, we can obtain unbiased estimates of β_1 , and β_2 since the error term and the independent variables $educ$ and exp are independent. Also, the OLS estimators are consistent, that is, converges in probability to the true parameter value as the sample size increases. However, this conclusion can be drastically altered. Suppose that instead of equation (3), the true model assumes that ability is also related with education e.g.

$$abil_i = \alpha_0 + \alpha_1 educ_i + \alpha_3 IQ_i + \varepsilon_i \quad (4)$$

Now, the model is,

$$\log(wage_i) = \delta_0 + \delta_1 educ_i + \beta_2 exp_i + \delta_3 IQ_i + v_i,$$

where δ_0 , δ_3 , v_i are defined as before, and $\delta_1 = \beta_1 + \alpha_1\beta_3$.

It can be shown that, as the sample size increases, the estimator of the return to education will tend to

$$\hat{\delta}_1 \rightarrow \beta_1 + \alpha_1\beta_3 \neq \beta_1,$$

which is different than the true parameter β_1 since $\alpha_1 \neq 0$. Thus, although we introduce a variable to account for the effect of the unobserved ability, we have an inconsistent estimator of the return to education parameter. For example, if years of education is positively related with ability ($\alpha_1 > 0$), and ability is positively related with earnings ($\beta_3 > 0$), the asymptotic bias is positive.

2 IV Estimation

We can consider replacing x with another variable, say z . The new variable, called instrumental variable, has to satisfy certain conditions¹:

1. The instrumental variable z has to be uncorrelated with the error term u , $E(zu) = 0$
2. The instrumental variable z has to be correlated with the endogenous variable x , $E(zx) \neq 0$

Consider the following linear regression model,

$$y = X_1\beta_1 + X_2\beta_2 + u = X\beta + u$$

where X_1 are exogenous variables (e.g. $\mathbb{E}(u|X_1) = 0$) and X_2 are endogenous -observed- variables (e.g. $\mathbb{E}(u|X_2) \neq 0$). Suppose also that there is a matrix of instruments Z . The idea is as follows: multiply each side of the equation by the matrix of instrumental variables Z ,

$$Z'y = Z'X\beta + Z'u$$

where Z may include X_1 as instrument for X_1 , and Z_2 as instrument for X_2 . Dividing by n and applying the Law of Large Numbers on $n^{-1}Z'u$, we get an expression for β that leads to the IV estimator. (Why?). The instrumental variable estimator is,

$$\hat{\beta}_{IV} = (Z'X)^{-1}Z'y$$

Under spherical errors, the variance of the estimator

$$V(\hat{\beta}_{IV}) = \hat{\sigma}_u^2(Z'X)^{-1}Z'Z(Z'X)^{-1}$$

Note that the IV estimator $\hat{\beta}_{IV}$ can be also obtained from the moment condition

$$Z'(y - X\beta) = 0,$$

which involve p equations and p unknowns.

The IV estimator is unbiased under assumptions 1 and 2.(Why?). Note that

$$E(\hat{\beta}_{IV}) = \beta,$$

¹Why the IV has to satisfy these two conditions? The first condition may be intuitive. The first condition requires that z and u are uncorrelated implying that if we are going to use another variable, at least, it has to fix the original problem, $\text{Cov}(x, u) \neq 0$. Could you explain the second condition?

because $E(Z'u) = 0$. Also it is consistent

$$\hat{\beta}_{IV} - \beta = (n^{-1}Z'X)^{-1}(n^{-1}Z'u) \rightarrow 0$$

provided that the limiting form of $Z'X$ is deterministic and nonsingular. (Conditions needed for asymptotic identification). The errors are asymptotically uncorrelated with the instruments, which follows from condition 1 by the law of large numbers. Note that this condition may hold even if condition 1 does not.

What are the instruments Z ? First since $X_1 \perp u$, X_1 can be (trivially) used as IV for itself. We then need additional instruments satisfying 1 and 2 to identify the model. The vector Z_2 are IV variables for X_2 , often denoted as excluded (from the model) exogenous variables.

The IV estimator has the following asymptotic representation,

$$\sqrt{n}(\hat{\beta}_{IV} - \beta) \rightsquigarrow \mathcal{N}(0, \sigma^2 Q_{zx}^{-1} Q_{zz} Q_{zx}^{-1})$$

where $Q_{zx} = \lim_{n \rightarrow \infty} n^{-1}Z'X$. If $Z = X$, the IV estimator is trivially the OLS estimator. Lastly, note that the relationship between Z and X in large samples place a key role for determining the asymptotic variance.

2.0.1 Example: IV Wald Estimator

Consider simple regression model

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

where $\mathbb{E}(u_i|x_i) \neq 0$. Our model's assumptions are (a) the error term has expected value zero, and (b) the correlation between the instrumental variable and the error term is zero. These population conditions impose condition over the data counterpart that can be stated as

$$E(u) = 0 \Rightarrow \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0 \quad (5)$$

$$E(zu) = 0 \Rightarrow \sum_{i=1}^n (z_i(y_i - \beta_0 - \beta_1 x_i)) = 0 \quad (6)$$

From the first equation, we obtain $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$. Replacing this expression in the second equation gives the Wald estimator,

$$\sum_{i=1}^n z_i(y_i - \bar{y}) = \hat{\beta}_1 \sum_{i=1}^n z_i(x_i - \bar{x})$$

The variable $z_i = \{0, 1\}$ is a dummy variable, therefore we can write,

$$\sum z_i(y_i - \bar{y}) = N_1(\bar{y}_1 - \bar{y}) = N_1\left(\frac{N_0\bar{y}_1 + N_1\bar{y}_1}{N} - \frac{N_0\bar{y}_0 + N_1\bar{y}_1}{N}\right) = \frac{N_1N_0}{N}(\bar{y}_1 - \bar{y}_0),$$

Consequently, we can write the estimator as,

$$\hat{\beta}_{Wald} = \frac{(\bar{y}_1 - \bar{y}_0)}{(\bar{x}_1 - \bar{x}_0)}$$

3 Generalized Instrumental Variable

Note that $(Z'X)$ invertible needs to have a Z matrix with p instruments. When the number of IVs is equal to the number of parameters to be estimated, we say that the model is exactly identified. Often the number of IVs is greater than the number of parameters, so we can consider the following matrix

$$W = P_z X$$

to achieve a transformation. The model is said to be overidentified, and in this case the moment condition is

$$W'(y - X\beta) = 0,$$

giving the generalized instrumental variable estimator,

$$\hat{\beta} = (W'X)^{-1}W'y$$

The generalized IV estimator is usually known as the Two Stage Least Square (2SLS) estimator,

$$\hat{\beta}_{2SLS} = (X'P_zX)^{-1}X'P_zy$$

which is based on two projections²:

1. Regress X on Z to obtain,

$$\hat{X} = Z(Z'Z)^{-1}Z'X = P_zX$$

²The main idea of the 2SLS estimator is the following. In a binary regression model consider that the independent variable is endogenous:

$$y_i = x_i'\beta + u_i; \quad x_i = z_i'\pi + v_i$$

The second equation is interesting since it contains two parts that provides the decomposition needed to solve the endogeneity in the main equation $\pi_0 + \pi_1 z_i$ is uncorrelated with u but v_i is correlated with u . The idea of 2SLS is to use only the first part, the part of the decomposition that it is independent of the error term. The problem is that $z_i'\pi$ is not observed, but it can be predicted using the simple OLS regression $\hat{x}_i = z_i'\hat{\pi}$.

It is important to see the components of the projection matrix $\hat{X} = [P_z X_1, P_z X_2]$. Recall that $Z = [X_1, Z_2]$, therefore,

$$\hat{X} = [X_1, \hat{X}_2]$$

2. Then regress y on $\hat{X}$, obtaining the 2SLS estimator $\hat{\beta}_{2SLS} = (\hat{X}'\hat{X})^{-1}\hat{X}'y$.

If the errors are homocedastic, the previous conditions (1)-(2) are satisfied, and the rank condition (as many IV's as independent variables) is satisfied, we have the following result.

Theorem 1. *The 2SLS estimator is asymptotically normally distributed with mean zero and variance matrix,*

$$\text{plim}_{n \rightarrow \infty} [n^{-1} \sigma^2 (X' P_z X)]^{-1}$$

Proof: Write the IV estimator as,

$$\hat{\beta} = \beta + (X' P_z X)^{-1} X' P_z u$$

Then, multiply each term by $\sqrt{n}$ and write it as,

$$\sqrt{n}(\hat{\beta} - \beta) = (n^{-1} X' P_z X)^{-1} (n^{-1/2} X' P_z u) + o_p(1)$$

The result follows passing n to ∞ and applying Slutsky theorem. ■ (Can you modify the result for the case of non-spherical errors?).

3.1 A Third Derivation

An interesting interpretation of the 2SLS is as follows. Define the residuals of X on Z ,

$$R = X - \hat{X} = X - P_z X = M_z X$$

Consider now the least square estimator β from a model that includes the residuals,

$$y = X\beta + R\gamma + u,$$

which is equal to

$$\hat{\beta}_{cv} = (X' M_R X)^{-1} X' M_R y$$

where cv stands for control variate. This estimator is identical to the 2SLS estimator because,

$$\begin{aligned} X' M_R &= X'(I - R(R'R)^{-1}R') = X'(I - (X - P_z X)(X' M_z X)^{-1}R') \\ &= X' - R' = X' - (X - P_z X)' = X' P_z \end{aligned}$$

Thus the “endogeneity bias” can be viewed as an omitted variable bias.

4 Tests

4.1 A misspecification test

Of course, in practical situations, we do not know whether the independent variables are correlated with the error term. We may use economic theory to guide us, but the real statistical question is simply: Is the endogeneity issue serious enough? For example, if the answer is no, we may ignore the issue and simply estimate the model using OLS.

The previous question can be examined with a test proposed by J. Hausman in 1978. The test about specification models in econometrics has a wide applicability to different models. In panel data, as it will be clear in future lectures, it helps to decide between the fixed and random effects models by testing the assumed hypothesis of independence between the individual specific effects and the independent variables.

In our cross-sectional setting, the null and alternative hypothesis may be formulated as,

$$H_0 : \mathbb{E}(X'u) = 0; \quad H_a : \mathbb{E}(X'u) \neq 0,$$

Under H_0 the OLS and IV estimators are consistent, though the OLS is efficient. Under H_a , the IV estimator is still consistent, but the OLS is inconsistent.

Therefore, a test based on the difference between these two estimators may indicate if the endogeneity affect the consistency of the OLS estimator. The idea is to evaluate whether the distance between the two estimators is asymptotically zero,

$$\hat{q} = \tilde{\beta} - \hat{\beta} \rightarrow 0$$

where $\tilde{\beta}$ is the IV estimator, $\hat{\beta}$ is the OLS estimator. What is the distribution of q ? We expect that q is asymptotically normal with asymptotic variance,

$$V(\tilde{\beta}) - V(\hat{\beta})$$

Lemma 1. Under H_0 , $V(\tilde{\beta} - \hat{\beta}) = V(\tilde{\beta}) - V(\hat{\beta})$

Proof: Write

$$\Delta = \tilde{\beta} - \hat{\beta} = [(X'P_zX)^{-1}X'P_z - (X'X)^{-1}X']u = [A - B]u$$

Then,

$$\mathbb{E}\Delta\Delta' = \mathbb{E}[(A - B)uu'(A' - B')] = \sigma_u^2[AA' - AB' - BA' + BB']$$

Note that

$$AB' = (X'P_zX)^{-1}X'P_zX(X'X)^{-1} = (X'X)^{-1} = BA' = BB',$$

so

$$\mathbb{E}\Delta\Delta' = \sigma^2[AA' - BB']$$

■

A natural test statistics is then,

$$h = \hat{q}'(V(\tilde{\beta}) - V(\hat{\beta}))^{-1}\hat{q}$$

which is asymptotically distributed as χ^2 . If h is low, say less than χ^2 critical value, we do not reject the null hypothesis. We conclude that the endogeneity does not affect the consistency of the OLS estimator. If h is high, say greater than χ^2 critical value, we reject the null hypothesis, and we conclude that endogeneity is a problem for the OLS estimator. Thus, we should use the IV estimator.

The Hausman specification test can be also used to evaluate two 2SLS estimators just by testing the exogeneity of the dubious IV variable. We can maintain the exogenous set of instruments and then we ask whether introducing the dubious IV variable is a good or bad idea.

4.2 Overidentifying Restrictions

Consider the regression model is well specified, $y = X\beta + u$. A candidate Z must satisfy,

$$E(u|Z) = 0 \text{ or } E(Z'u) = 0$$

Note that in the case of just indentified models, we have that

$$Z'\hat{u} = 0,$$

so we cannot test the previous conditions. However, if we have more instruments (say k) than endogenous variables (say q), we can consider the following test,

$$T = \hat{\sigma}^{-2}\hat{u}'Z(Z'Z)^{-1}Z'\hat{u} = \hat{\sigma}^{-2}\hat{u}'P_Z\hat{u} \sim \chi^2_{k-q},$$

under the null hypothesis. If we reject the null, this should be interpreted as evidence that instruments are endogenous. Let me present you a simple way to implement the test:

1. Estimate the structural equation by 2SLS to obtain the residuals, $\hat{u}_i$
2. Regress $\hat{u}_i$ on the exogenous variables, the IVs $z_1, \dots, z_k$, and the included covariates $x_q, \dots, x_p$,
3. With the R^2 of the last regression, construct the test statistic nR^2 to evaluate the null hypothesis.

5 Weak Instruments

Let me now say a word about the consequences of choosing an “weak” instrument, that is, a variable that it does not satisfy either conditions (1) or (2). Suppose, for example, that z violates (1), that is, the instrumental variable is not independent of the error term. Considering a model with one (endogenous) regressor,

$$y = \beta x + u$$

it can be shown that,

$$\hat{\beta} \rightarrow \beta + \frac{\text{Cor}(z, u)}{\text{Cor}(z, x)} \frac{\sigma_u}{\sigma_x}$$

where $\text{Cor}(\cdot)$ means correlation.

This is an interesting formula. Even if the correlation between z and u is small, the asymptotic bias can be very large if (a) z , and x are barely correlated, and (b) the noise-to-signal ratio (σ_u/σ_x) is large.

If we compare the asymptotic bias of the poor instrumental variable estimator with the OLS estimator in the case that x is not exogenous,

$$\hat{\beta} \rightarrow \beta + \frac{\text{Cov}(x, u)}{V(x)} = \beta + \frac{\sigma_x \sigma_u \text{Cor}(x, u)}{\sigma_x^2} = \beta + \text{Cor}(x, u) \frac{\sigma_u}{\sigma_x}$$

we may conclude that the OLS estimator is preferred to the 2SLS estimator if

$$\frac{\text{Cor}(z, u)}{\text{Cor}(z, x)} > \text{Cor}(x, u)$$

It is not surprising then that it is often stated that a weak IV can be detected using the first stage regression (e.g., simply looking at the F test or R^2). Standard practice is to check if the instruments are correlated with the endogenous variable by running the endogenous on the IV's and the exogenous variables. An F test on the coefficients on the iv's provides a measure of the information content of the IV's.³

Unfortunately, the problem of weak IV's is typically more complex since it includes other parameters. Consider for instance,

$$\begin{aligned} y_1 &= \beta y_2 + u \\ y_2 &= z' \pi + v, \end{aligned}$$

³A rule of thumb, taken from Stock and Watson (2003), suggests that 10 is the value of the F -statistic that divides weak and valid instruments. For example, if the F statistic exceeds 10, we reject the null hypothesis concluding that the z s are valid instruments for our endogenous variable.

where z is a k -dimensional vector of instruments and $E(uv) = \sigma_{uv}$. The 2SLS estimator is:

$$\hat{\beta} = \frac{y_2' P_z y_1}{y_2' P_z y_2} = \beta + \frac{y_2' P_z u}{y_2' P_z y_2}.$$

Let $q_2 = y_2' P_z y_2 / n$ and note that $y_2' P_z u = \text{tr} y_2' P_z u = \text{tr} P_z y_2' u = \text{tr}(I_k) y_2' u$. Then, one can express the bias of the 2SLS estimator as:

$$\hat{\beta} - \beta \approx \frac{k}{n} \frac{\sigma_{uv}}{q_2} \approx \frac{k}{n R^2} \frac{\sigma_{uv}}{\sigma_{y_2}^2}.$$

This implies that k , the coefficient of determination of the first stage regression R^2 and the correlation between the errors in the equations σ_{uv} play a role in determining the bias of the 2SLS in small samples. Note that as $n \rightarrow \infty$, if k is fixed, we tend to have (asymptotically) zero bias. The literature offers tests and biased-corrected estimators (see e.g., Stock 2002, Hahn and Hausman 2003).

Another problem is related to inference when the correlation between the endogenous variable and the instrument is very small. Again, consider the following model,

$$\begin{aligned} y_1 &= \beta y_2 + u \\ y_2 &= z\pi + v, \end{aligned}$$

where now $k = 1$. In other words, the model is exactly identified and we instrument the endogenous using one instrument. Recall that $\pi = E(z_i y_{2i}) / E(z_i^2)$ and consider that $\pi = \gamma / \sqrt{n}$. In this case,

$$\hat{\beta} - \beta = \frac{\frac{1}{\sqrt{n}} \sum_i z_i u_i}{\frac{1}{\sqrt{n}} \sum_i z_i y_{2i}} \rightsquigarrow \frac{\mathcal{N}(0, V_u)}{Q_z \gamma + \mathcal{N}(0, V_v)},$$

distributed as non-normal for $\gamma \neq 0$ and Cauchy as $\gamma = 0$. Therefore, the IV estimator converges in probability to a non-degenerate random variable. Its distribution is non-normal and consequently inference procedures are inconsistent too (see e.g. Choi and Phillips 1992)

6 IV estimation in non-linear models

Measurement error (ME) in the independent variables causes:

1. bias and inconsistency in parameter estimation for econometric models, and
2. loss of power for detecting stochastic significant relationship between the variables of

interest

The techniques to address the problem of measurement error can be classified in two groups: (a) methods for linear error-in-variables models and (b) methods for non-linear error-in-variables models.

6.1 Classical Errors

This refers to a situation where the true latent variable is uncorrelated with the measurement error. Consider,

$$Y = \alpha + \beta X^* + \epsilon \quad (7)$$

$$X = X^* + u \quad (8)$$

and therefore,

$$Y = \alpha + \beta X + (\epsilon - \beta u) = \alpha + \beta X + v, \quad (9)$$

where $v = \epsilon - \beta u$. Recall that the biased and inconsistent estimator for the slope is,

$$\hat{\beta} = \frac{\sum (X_i - \bar{X}) Y_i}{\sum (X_i - \bar{X})^2} \quad (10)$$

and the intercept is,

$$\hat{\alpha} = \bar{Y} - \hat{\beta} \bar{X} \quad (11)$$

Note that,

$$\hat{\alpha} \rightarrow \mathbb{E}(\alpha + \beta X^* + \epsilon) - \frac{\beta \sigma_{X^*}^2}{\sigma_{X^*}^2 + \sigma_u^2} \mathbb{E}(X^* + u) \quad (12)$$

$$= \alpha + \frac{\beta \sigma_{X^*}^2}{\sigma_{X^*}^2 + \sigma_u^2} \mathbb{E}(X^*) \quad (13)$$

Remark 1. It is interesting to run a regression for X on Y to find the bounds for the true coefficients α and β . Write,

$$X = -\frac{\alpha}{\beta} + \frac{1}{\beta} Y - \frac{v}{\beta} \quad (14)$$

and therefore,

$$\tilde{b} = \frac{\sum_{i=1}^n (X_i - \bar{X}) Y_i}{\sum_{i=1}^n (Y_i - \bar{Y})^2}, \quad (15)$$

$$\tilde{\alpha} = \bar{Y} - \tilde{b} \bar{X} \quad (16)$$

where $\tilde{\beta} = \tilde{b}^{-1}$. Now we write that

$$\tilde{\alpha} \rightarrow \mathbb{E}(\alpha + \beta X^* + \epsilon) - (\beta + \frac{\sigma_u^2}{\beta \sigma_{X^*}^2}) \mathbb{E}X^* = \alpha - \frac{\sigma_u^2}{\beta \sigma_{X^*}^2} \mathbb{E}X^* \quad (17)$$

Example 1. Note that plim of $\tilde{b} = \text{Cov}(x, y) / \text{Var}(y)$. It is possible to show that

$$\text{plim}\tilde{\beta} = \beta + \sigma_u^2 / \beta \sigma_{X^*}^2$$

As previously discussed, the model with classical errors can be seen as a problem of endogenous regressors. Therefore, IV turns to be a popular approach for estimating such models.

Hausman-type tests can also be used in this type of models. A valid IV could have the form $Z = X^* + v_Z$, with v_Z being an independent error: $\mathbb{E}(Z(\epsilon, u)) = 0$.

Example 2. Consider the previous model, and assume that $\alpha = 0, \beta = 1, \sigma_\epsilon^2 = 0.25, \sigma_u^2 = 1$, and $X^* \in \mathcal{N}(0, 1)$. We can illustrate the attenuation bias using the following two graphs.

Example 3. When we have many columns in X and we observe only one covariate with measurement error, the formula for the attenuation bias is

$$\hat{\beta} \rightarrow \beta(1 - \frac{\sigma_u^2}{\sigma_{X^*}^2(1 - R_{X^*, Z}^2) + \sigma_u^2}) \quad (18)$$

where R^2 is the true R^2 .

Example 4. Suppose we observe repeated measurements for individual over time. For instance, assume

$$Y_{it} = \alpha + \beta X_{it}^* + \mu_i + \epsilon_{it} \quad (19)$$

$$X_{it} = X_{it}^* + u_{it} \quad (20)$$

To remove that unobserved heterogeneity, we simply obtain first differences,

$$\Delta Y_{it} = \beta \Delta X_{it}^* + \Delta \epsilon_{it} \quad (21)$$

$$= \beta \Delta X_{it} + \Delta \epsilon_{it} - \beta \Delta u_{it} \quad (22)$$

$$= \beta \Delta X_{it} + \Delta v_{it} \quad (23)$$

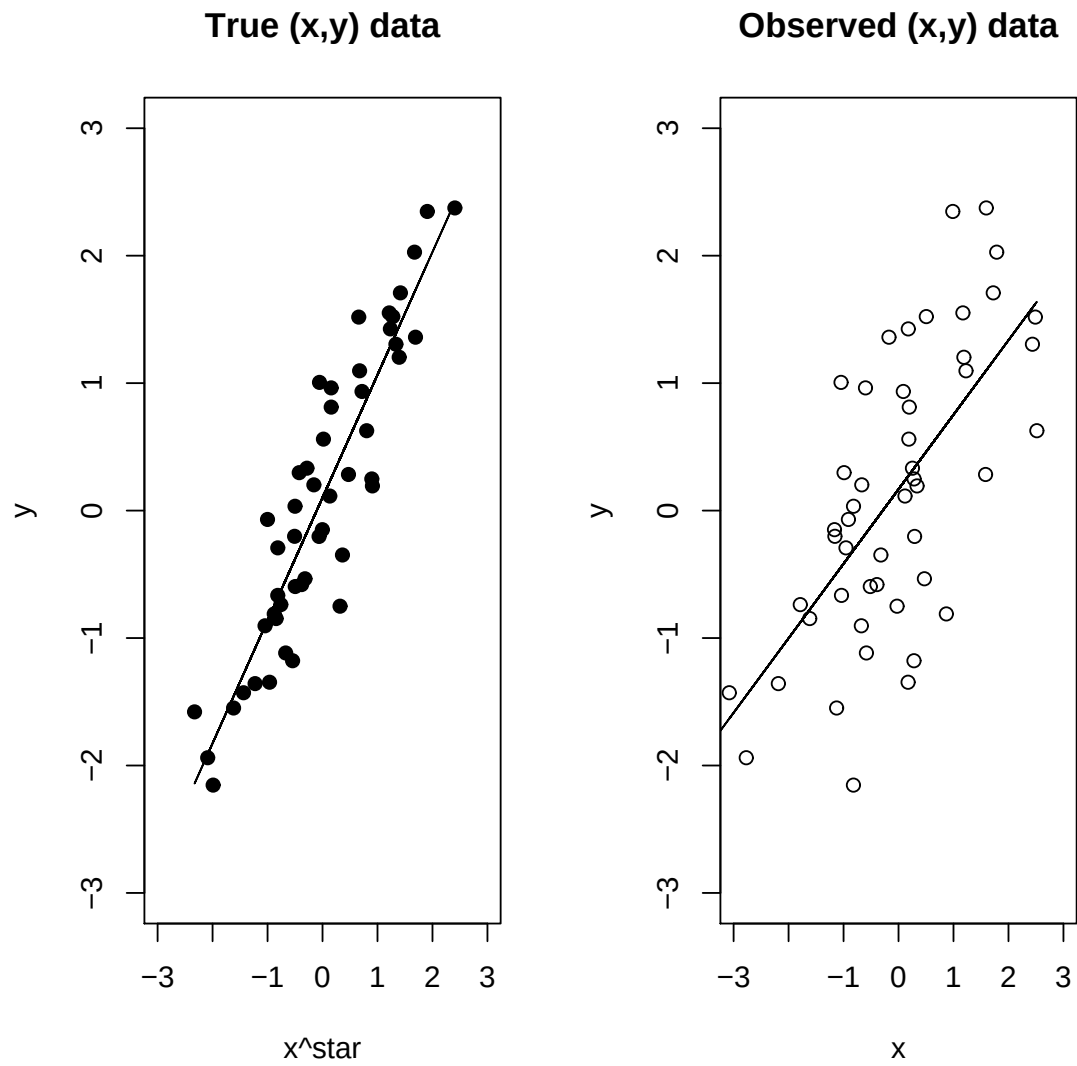


Figure 1: A simple example of the attenuation bias

Therefore,

$$\hat{\beta} = \sum_{i=1}^N \Delta X_{it} \Delta Y_{it} / \sum_{i=1}^N \Delta X_{it}^2 \quad (24)$$

$$= \beta + \frac{1/N \sum_{i=1}^N \Delta X_{it} (\Delta \epsilon_{it} - \beta \Delta u_{it})}{1/N \sum_{i=1}^N \Delta X_{it}^2} \quad (25)$$

Note that,

$$\mathbb{E}(\Delta X_{it})^2 = \mathbb{E}((X_{it}^* + u_{it}) - (X_{it-1}^* + u_{it-1}))^2 \quad (26)$$

$$= \mathbb{E}((X_{it}^* + u_{it})^2 - 2(X_{it}^* + u_{it})(X_{it-1}^* + u_{it-1}) + (X_{it-1}^* + u_{it-1})^2) \quad (27)$$

$$= \sigma_{X^*}^2 + \sigma_u^2 - 2Cov(X_{it}^*, X_{it-1}^*) + \sigma_{X^*}^2 + \sigma_u^2 \quad (28)$$

$$= 2(1 - \rho)\sigma_{X^*}^2 + 2\sigma_u^2 \quad (29)$$

where $\rho = Cor(X_{it}^*, X_{it-1}^*)$. Similarly, it can be shown that

$$\hat{\beta} \rightarrow \beta - \frac{2\beta\sigma_u^2}{2(1 - \rho)\sigma_{X^*}^2 + 2\sigma_u^2} \quad (30)$$

$$= \frac{\beta((1 - \rho)\sigma_{X^*}^2 + \sigma_u^2) - \beta\sigma_u^2}{(1 - \rho)\sigma_{X^*}^2 + \sigma_u^2} \quad (31)$$

$$= \frac{\beta(1 - \rho)\sigma_{X^*}^2}{(1 - \rho)\sigma_{X^*}^2 + \sigma_u^2} \quad (32)$$

6.2 Non-linear Models

It is well known that standard IV assumptions are not enough for identification of nonlinear error-in-variables models. Amemiya (1985) showed that the classical instrumental variable estimator may be inconsistent without additional structure. The literature offers a variety of these restrictions, e.g. from known distributions of measurement errors to stronger forms of IV's.

To illustrate the issues behind classical IV in non-linear settings, let's consider one example introduced in page 912 of Cameron and Trivedi (2005). Consider,

$$Y = \alpha + \beta f(X^*) + \epsilon \quad (33)$$

$$X = X^* + u, \quad (34)$$

where $f(\cdot)$ is a smooth function of a latent variable, X^* . Taylor expansion of $X^* = (X - u)$

around X yields'

$$f(X^*) = f(X - u) = f(X) + \frac{\partial f}{\partial X}(X^* - X) + \frac{1}{2!} \frac{\partial^2 f}{\partial X^2}(X^* - X)^2 \quad (35)$$

$$+ \dots + \frac{1}{n!} \frac{\partial^n f}{\partial X^n}(X^* - X)^n + \dots \quad (36)$$

$$= f(X) - f'u + \frac{1}{2}f''(-u)^2 + \dots + \frac{1}{n!}f^n(-u)^n + \dots \quad (37)$$

Suppose for simplicity that $f = \gamma X + X^2$ is quadratic. In this case, the approximation is exact (e.g., we do not have a reminder term R). Note that $f' = 2X + \gamma$ and $f'' = 2$. Replacing we have that,

$$Y = \alpha + \beta((\gamma X + X^2) - (2X + \gamma)u + \frac{1}{2} \cdot 2u^2) + \epsilon \quad (38)$$

$$= \alpha + \beta\gamma X + \beta X^2 + (\epsilon - 2\beta Xu - \beta\gamma u + \beta u^2). \quad (39)$$

Therefore, IV needs to satisfy:

1. $\mathbb{E}(Z_1 \cdot X) \neq 0$ and $\mathbb{E}(Z_2 \cdot X^2) \neq 0$
2. $\mathbb{E}(Z_j \cdot (\epsilon - 2\beta Xu - \beta\gamma u + \beta u^2)) = 0$,

which is obviously not equivalent to $\mathbb{E}(Z_j \cdot \epsilon) = \mathbb{E}(Z_j \cdot u) = 0$.